

Question Paper Preview

Notations :

- 1.Options shown in **green** color and with  icon are correct.
- 2.Options shown in **red** color and with  icon are incorrect.

Change Font Color :	No
Change Background Color :	No
Change Theme :	No
Help Button :	No
Show Reports :	No
Show Progress Bar :	No

Computer System Design

Section Id :	640653124862
Number of Questions :	21
Section Marks :	50
Maximum Instruction Time :	0

Question Number : 1 Question Id : 6406531867614 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : COMPUTER SYSTEM DESIGN (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406536048680.  YES

6406536048681.  NO

Question Number : 2 Question Id : 6406531867615 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

A multiplexer is a combinational circuit that selects one input from multiple inputs and forwards it to a single output.

Options :

6406536048682. ✓ True

6406536048683. ✗ False

Question Number : 3 Question Id : 6406531867616 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

A JK flip-flop cannot store information without a continuous clock input.

Options :

6406536048684. ✗ True

6406536048685. ✓ False

Question Number : 4 Question Id : 6406531867617 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

In a 4-bit system, the two's complement of a binary number 1010 can be represented as -6.

Options :

6406536048686. ✓ True

6406536048687. ✗ False

Question Number : 5 Question Id : 6406531867618 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

A decoder performs the inverse operation of a multiplexer.

Options :

6406536048688. ✓ True

6406536048689. ✗ False

Question Number : 6 Question Id : 6406531867619 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

According to Moore's Law, the size of transistors on an integrated circuit doubles approximately every two years.

Options :

6406536048690. ✖ True

6406536048691. ✔ False

Question Number : 7 Question Id : 6406531867620 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

The solutions of the quadratic equation

$$x^2 - 11_b x + 22_b = 0,$$

are $x = 3$ and $x = 6$. What is the base b of the numbers?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

8

Question Number : 8 Question Id : 6406531867621 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

The circuit diagram requires the output

$$f = (x + \overline{y})zw + \overline{z}w$$

The gates GT1 and GT2 should be.

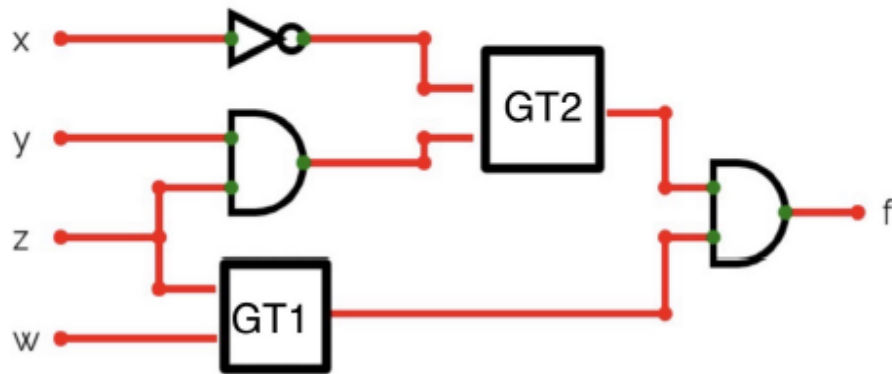


Figure 1: Circuit Diagram

Options :

6406536048693. ✖ OR, NOR

6406536048694. ✔ OR, NAND

6406536048695. ✖ AND, NAND

6406536048696. ✖ AND, NOR

6406536048697. ✖ None of these

Question Number : 9 Question Id : 6406531867622 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

What is the minimum number of 2-input XNOR gates required to implement the circuit for minimized SOP expression of the function.

$$f(x, y, z) = \sum m(0, 2, 5, 7)$$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1

Question Number : 10 Question Id : 6406531867623 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Given that NOT, 2-input AND/OR gates have delays of 1 ns, what will be the maximum delay of a 2x1 multiplexer circuit?

Note: if the answer is 1 ns, write 1.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

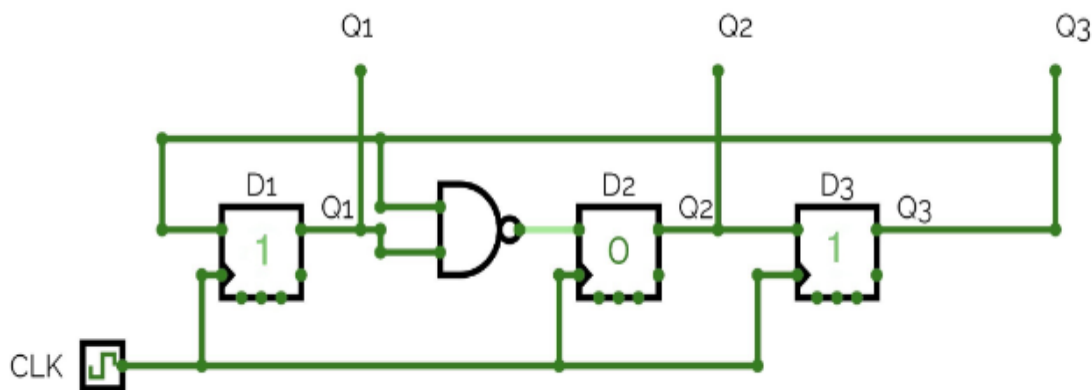
3

Question Number : 11 Question Id : 6406531867624 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider the circuit consisting of three D flipflops and one NAND gate as shown below. The initial states are $Q_1=1$, $Q_2=0$, and $Q_3=1$. What is the output after (Q_1, Q_2, Q_3) after 4 clock pulse?



Note: If answer if $Q_1=0$, $Q_2=0$, $Q_3=0$, then write 000

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

111

Question Number : 12 Question Id : 6406531867625 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

A processor uses the following control word format:

Opcode	Src1	Src2	Dest
10	10101	01110	00011

How many registers can be addressed in the register bank?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

32

Question Number : 13 Question Id : 6406531867626 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Using 2's complement method, evaluate $A - B$ where $A = 80$ and $B = 150$. Give the answer in 8-bit binary format.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

10111010

Question Number : 14 Question Id : 6406531867627 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Choose the correct truth table according to the given Boolean expression:

$$F(x, y, z) = \overline{xyz} + \overline{xy} + x\overline{z}$$

Options :

x	y	z	F
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

6406536048703. ✖

6406536048704. ✖ None of these

x	y	z	F
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

6406536048705. ✔

6406536048706. ✖

x	y	z	F
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Question Number : 15 Question Id : 6406531867628 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a common bus architecture containing memory and three registers: A, B, and C. The control word format is shown below.

Memory		Register A		Register B		Register C	
Read	Write	Input	Output	Input	Output	Input	Output

What would be the control word if you intend to perform a register transfer from register A to register B?

Note: Provide your answer in hexadecimal format (e.g., 0x3A).

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

0x18

Question Number : 16 Question Id : 6406531867629 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

A modern RISC processor contains 64 registers in its general-purpose register file. The Arithmetic Logic Unit (ALU) of the processor is capable of executing 16 distinct operations, which include standard arithmetic and logical functions.

What is the *minimum* number of bits required by a *three-address* instruction to perform a bitwise **OR** operation between two source operands located in the register file, and store the computational result back into a designated destination register?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

22

Question Number : 17 Question Id : 6406531867630 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Suppose we want to execute a data movement instruction, where we move the data present in register R_{25} to register R_{18} .

Instruction: $\text{MOV } R_d, R_r$

R_d : Destination register, R_r : Source register

Operands: $0 \leq d \leq 31, 0 \leq r \leq 31$

Instruction Format (16-bit):

0010	11rd	drdr	rrdd
------	------	------	------

What will be the hexadecimal machine code for the instruction? Note: Enter the value in hexadecimal format (e.g. 0x24E8).

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

0x2F46

Question Number : 18 Question Id : 6406531867631 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a gated D-Latch constructed using NAND gates. The initial input (at 0 ns) is $D = 1$, and the write enable is high. The write enable toggles between high and low every 5 ns. The input D changes from 0 to 1 and from 1 to 0 every 6 ns. What will be the output Q after 16 ns?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1

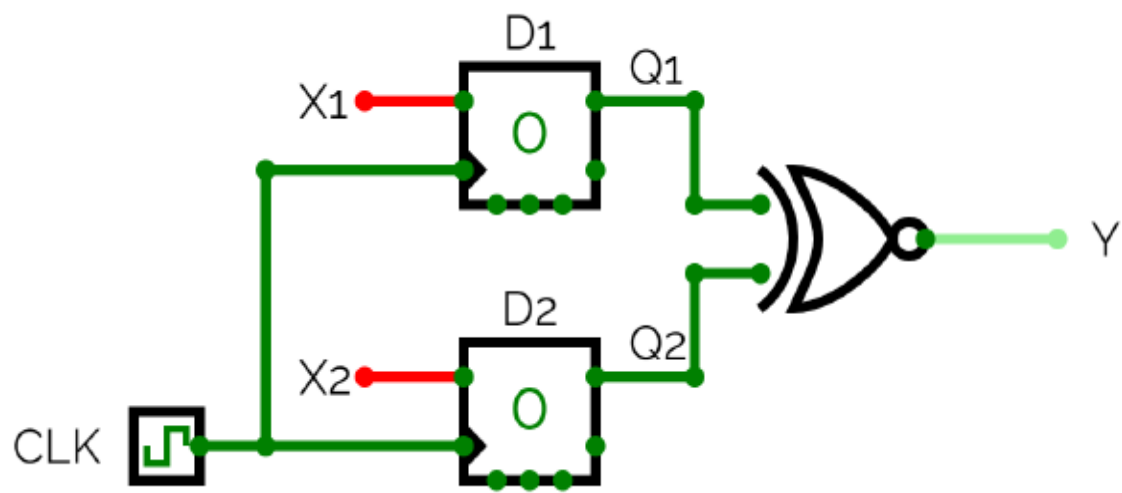
Question Number : 19 Question Id : 6406531867632 Question Type : MCQ

Correct Marks : 3

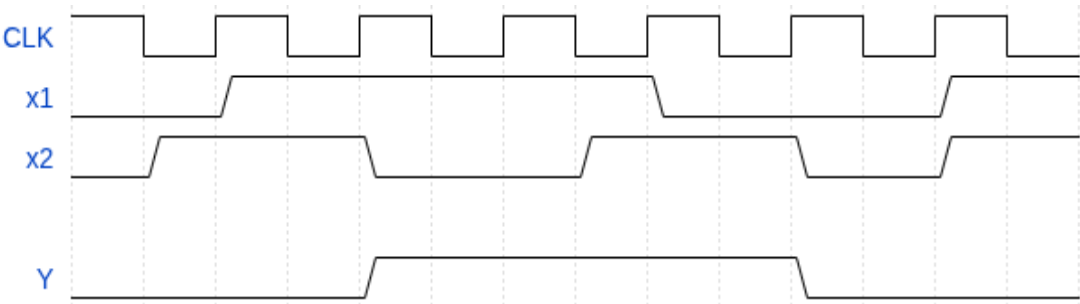
Question Label : Multiple Choice Question

In the circuit shown below, D1 and D2 are positive edge triggered D flipflops, x1 and x2 are inputs of D1 and D2 flipflop.

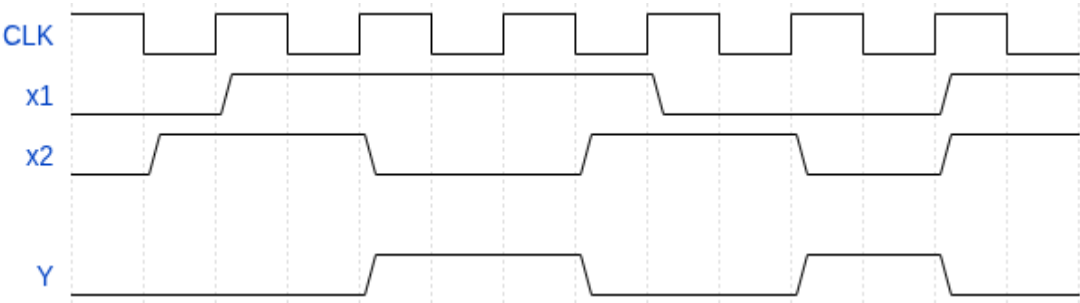
Choose the correct timing diagram of the output Y.



Options :

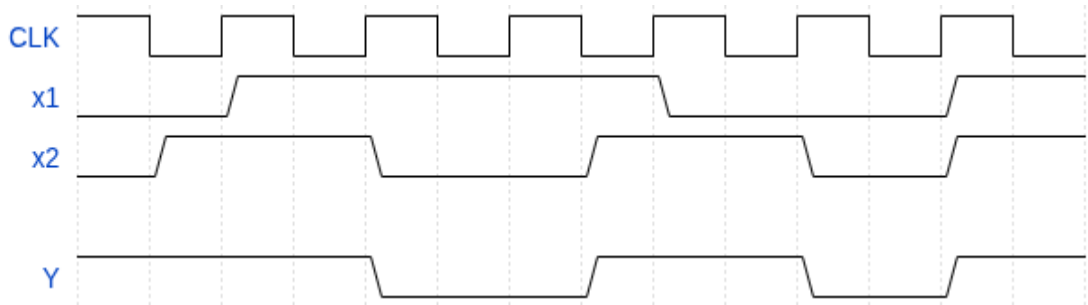
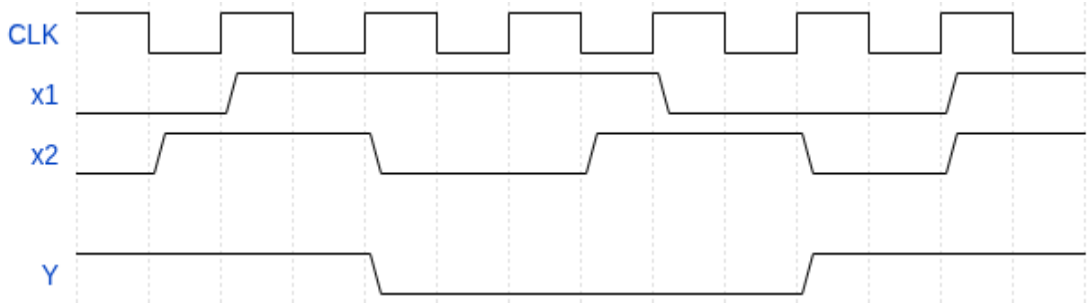


6406536048711. ✖



6406536048712. ✖

6406536048713. ✔



6406536048714. ✖

6406536048715. ✖ None of these

4